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Kinetic characteristics and predictive models of methylmercury production in paddy soils☆

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ABSTRACT

Understanding the mercury (Hg) methylation process is important for the management of paddy soils contaminated by Hg. In this work, samples of eighteen paddy soils with varying soil properties were spiked with inorganic Hg and subjected to a 90 d flooding period. Soil pH and redox potential (Eh) were measured in situ at intervals, and soils were sampled for the analysis of methylmercury (MeHg). The Hg methylation efficiency increased with flooding time and reached a relatively steady state at 30 d of incubation, ranging from 0.08% to 2.52%, and was significantly correlated with the in situ soil pH and Eh. The Elovich equation could adequately describe the kinetic production of MeHg. MeHg production was well predicted by the in situ soil pH and Eh of flooded soils, in addition to the organic matter content of air-dried soil samples and flooding time. The two predictive models explained 78% and 68% of the variability of the Hg methylation efficiency. The results suggested that the methylation of inorganic Hg in paddy soils after flooding can be predicted as a function of routinely measured soil properties and flooding time, a correlation that can be utilized to improve understanding of the extent of Hg methylation and the management of Hg-contaminated paddy soils.

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1. Introduction

Methylmercury (MeHg) is the most toxic chemical form of mercury (Hg) (Clarkson, 1998, Clarkson et al., 2003). The Hg methylation process mainly occurs in anaerobic environments with microorganisms that include sulfate-reducing bacteria (SRB), iron-reducing bacteria (FeRB), and methanogens (Compeau and Bartha, 1985; Kerin et al., 2006; Bravo et al., 2018). The Hg methylation efficiency is affected by microbial activity and Hg bioavailability, both of which are influenced by soil properties. Anaerobic paddy soil environments are beneficial for Hg methylation (Duvil et al., 2018; Rothenberg et al., 2016; Wang et al., 2014); however, the Hg methylation process in paddy soil is not well understood.

It has been demonstrated that soil properties can affect the availability of inorganic Hg and the activity of methylation microorganisms (Wang et al., 2016; Yin et al., 2018; Zhao et al., 2019). The main soil physicochemical properties that affect Hg methylation in paddy soils have been investigated for organic matter (Lambertsson and Nilsson, 2006; Ma et al., 2015), redox potential (Eh) (DeLaune et al., 2004; Garcia-Ordiales et al., 2018), sulfate (Gilmour et al., 2018) and selenium (Wang et al., 2016). Rothenberg and Feng (2012) suggested that Fe(III) reduction increased electron acceptors for sulfate-reducing bacteria under anaerobic conditions. Li et al. (2019) indicated that sulfate amendment could enhance Hg mobility and methylation in a mining-contaminated paddy field. The effects of different doses on these properties were investigated in previous studies but did not reflect the interactions between these soil properties. Furthermore, due to the limitations of soil type and geographic restriction in some studies (Gilmour et al., 2018; Qiu et al., 2012), previously identified influential factors might lack broader applicability.

It has been reported that the MeHg concentration of the soil (pH = 7.92, OM content 16.92 mg/kg, THg 0.082 mg/kg) in the water-level-fluctuating zone of the Three Gorges Reservoir, China increased with flooding time and reached equilibrium after flooding for 30 d in a 40-d laboratory incubation experiment (Wang et al., 2015). However, for the majority of paddy soils, the dynamic characteristics and equilibrium time of MeHg production are

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still unknown. A predictive model of MeHg production is necessary to demonstrate the major controlling factors and the extent of Hg methylation. The model will also quantitatively assess the environmental risks associated with Hg-contaminated paddy soils.

Therefore, the aims of the present study were (i) to investigate the kinetic characteristics of the methylation process of exogenous Hg and (ii) to identify the major controlling factors by developing predictive models of MeHg production using a wide range of Chinese paddy soils.

2. Materials and methods

2.1. Soil description and experimental design

Eighteen paddy soils with various soil properties were sampled from major rice production areas in China. These soils were sampled from the surface of paddy fields (0–20 cm depth) and were air-dried, homogenized, and sieved to pass through a 2 mm sieve. Soil physicochemical properties were measured according to Lu (2000), including the soil pH (1:2.5 soil to water ratio), cation exchange capacity (CEC) (1 M ammonium acetate leaching method at pH 7.0), organic matter (K_2CrO_4:H_2SO_4 oil bath heating), sulfate (1:10 soil to water ratio and measured by ion chromatography), clay content (the sieve-pipette method), and free Fe/Mn/Al (DCB method). These physiochemical properties are listed in Table S1.

One kilogram of each soil type was spiked with Hg(II) (5 mg/kg) using $HgCl_2$ solution to ensure sufficient MeHg production (Peng et al., 2012) and placed into a glass beaker; each soil type was evaluated in triplicate. All beakers were incubated in the dark at 25 °C for 90 d. Deionized water was added to form a 2 cm water layer on the soil surface. The soil pH and Eh were measured in situ at time intervals of 1, 3, 7, 14, 30, 60, and 90 d during the incubation period. In the meantime, soil samples were collected to determine the MeHg concentration.

During the incubation period, soil pH and Eh were measured in situ using a FieldScout pH 400 m (Spectrum Technologies Inc., USA) and a redox potentiometer (CP-0701-3, Nanjing Kuake Co. Ltd., China). The total Hg (THg) concentration of soil was determined according to Qian et al. (2017). Soils were digested by (1 + 1) aqua regia in 100 °C water bath. Then, the digested solutions were measured by a cold vapor atomic fluorescence spectrometer (CVAFS, AF-610D2, Beijing Rayleigh Analytical Instrument, China). The total selenium in soil was digested and measured using the same method as THg (Martens and Suarez, 1996). The MeHg in soil was measured according to Gagnon et al. (1997): soil samples were digested by 25% KOH/methanol (w/w) and shaken in an incubation shaker at 65 °C for 4 h; then, the digested samples were tested with GC-CVAFS (cold vapor atomic fluorescence spectrometry, Brooks Rand, USA) according to USEPA method 1630.

For quality control, standard materials (GBW07407 for THg and Se of soil, ERM-cc580 for MeHg of soil) were included in the analysis. Analysis of the standard material yielded THg, Se and MeHg concentrations of 716 ± 11 μg/kg, 300 ± 12 μg/kg and 73 ± 5.7 μg/kg, respectively, which were close to the certified values 740 ± 110 μg/kg, 320 ± 9 μg/kg and 75.5 ± 4 μg/kg. The LOD and LOQ of CVAFS for total Hg determination and GC-CVAFS for MeHg determination were 2.0 μg/kg and 0.2 pg, respectively.

The abundance of *hgc*A in the air-dried soils prior to incubation was quantified using a primer pair (hgcA4F; 5′-GGNRTYAAYRTCTGGTGYGC-3'; hgcA4R; 5′-CGCATYTCCTTYTY-BACNCC-3′) (Liu et al., 2014). The total microbial DNA was extracted with a FastDNA SPIN kit for soil (MP Biomedicals, Solon, OH) following the manufacturer's instructions. Real-time quantitative PCR (qPCR) was performed in triplicate with a real-time fluorescence quantitative PCR instrument (TIB8600, Triplex International Biosciences, China).

2.2. Data analysis

The Hg methylation efficiency (k_{ME}) was calculated by Eq. (1):

$$k_{ME} = \frac{MeHg_{soil}}{THg_{soil}} \quad (1)$$

where $MeHg_{soil}$ is the MeHg concentration in soil, and THg_{soil} is the total Hg concentration in soil.

Two normal equations were selected as the kinetic equations to describe the Hg methylation process in this study (Wang et al., 2015). One of the equations is the Elovich equation:

$$C_t = a + b\ln t \quad (2)$$

Another equation is the two-constant equation:

$$\ln C_t = a + b\ln t \quad (3)$$

where t is flooding time (d), C_t is the soil MeHg concentration after flooding for t days (μg/kg), a and b are two constants and the value of b represents the speed of Hg methylation. The data were fitted with the two equations by Sigmaplot 10.0.

SPSS 19.0 was used to analyze the correlation between the Hg methylation efficiency and soil properties. The predictive model was developed using Solver in Microsoft Excel 2013. The parameters of the predictive model were optimized by minimizing the sum of squared residuals (Zhang et al., 2017).

3. Results and discussion

3.1. Characteristics of MeHg production in soils

The MeHg concentrations of soils during 1, 3, 7, 14, 30, 60, and 90 d of incubation were measured, and the k_{ME} values of these soils were calculated (Fig. 1). The k_{ME} of all soils increased with flooding time and reached a relatively steady state on the 30th d of incubation. The final Hg methylation efficiency in these soils ranged from 0.08% to 2.31%, with an average of 0.82%. The change tendency of methylation efficiency in the eighteen paddy soils was in line

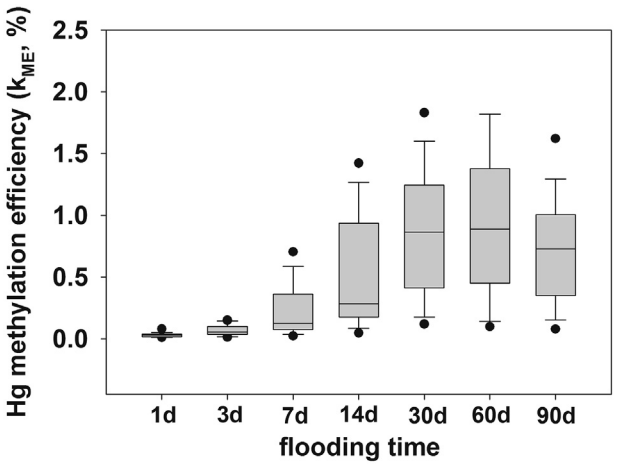


Fig. 1. Hg methylation efficiency of the eighteen paddy soil samples throughout the flooding duration. The box plots show the medians (solid line within boxes), 25th and 75th percentiles as horizontal bars, 10th and 90th percentiles as error bars and 5th and 95th percentiles as circles.

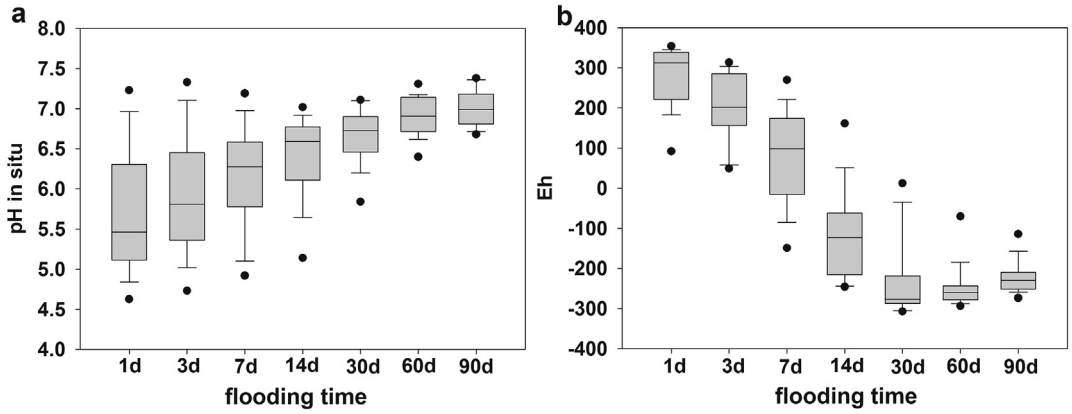


Fig. 2. The change tendency of the in situ pH and Eh for the eighteen soil samples. The box plots show the medians (solid line within boxes), 25th and 75th percentiles as horizontal bars, 10th and 90th percentiles as error bars and 5th and 95th percentiles as circles.

with those of previous studies. Zheng et al. (2017) suggested that the Hg methylation efficiency during 42 d of flooding incubation ranged from 0.43% to 0.76% for soil from Xiangyin, Hunan Province, China. Wang et al. (2015) suggested that the Hg methylation efficiency ranged from 0.60% to 1.72% in the water-level-fluctuating zone of the Three Gorges Reservoir in China. The Hg methylation efficiency ranged from 0.059% to 2.57% after 26 d of flooding incubation for eight soils from China spiked with 5 mg/kg inorganic Hg (Zhu et al., 2015). These data fell within the range of this study.

The in situ pH (Fig. 2a) and Eh (Fig. 2b) were found to have similar change tendencies and equilibrium times with the Hg methylation process, indicating that the characteristics of MeHg production were closely related to the observed changes in soil pH and Eh. There was a significant negative correlation between k_{ME} and Eh ($R^2 = 0.785$, $P < 0.001$) (Fig. 3a). On one hand, when the Eh decreased to −200 mV, the anaerobic environment might benefit the activity of methylation bacteria. On the other hand, lower Eh may increase Hg bioavailability because high-molecular-weight species of organic matter (OM) can convert into dissolved organic matter (DOM), and then Hg(II) is released and more easily methylated (Yang et al., 2007; Wang et al., 2017). In addition, Fe and Mn oxides would be reduced and dissolved to Fe(II) and Mn(II), causing the Hg(II) adsorbed to the Fe and Mn oxides to be released and promote the methylation process (Guo et al., 2016).

The in situ soil pH was found to be positively correlated with k_{ME} ($R^2 = 0.472$, $P < 0.001$) (Fig. 3b), indicating that k_{ME} has a similar change tendency with the in situ soil pH. It is generally considered that a lower soil pH can facilitate MeHg production. In fact, soils with lower pH values release more H^+, which increases bacterial uptake of Hg (II) (Kelly et al., 2003) and makes the change in pH after flooding more pronounced for these soils. Because soils with lower pH values release more H+, which increases the net intracellular accumulation of Hg(II), uptake into cells is enhanced (Golding et al., 2008).

The characteristics of MeHg production in soil were fitted by kinetic equations. Both the Elovich and two-constant equations were used to fit the MeHg concentrations in soil samples. The b values of these equations could be used to represent the rate of Hg methylation with time (Table S2). Soils with higher MeHg production ability had higher b values. According to Table S2, the Elovich equation may describe the kinetic characteristics of MeHg production better than the two-constant equation. The results of correlation analysis between soil properties and b values of the two equations indicated that OM content had a significant positive correlation with the b value of the Elovich equation (Table 1). The effect of OM content on k_{ME} includes three aspects: the OM content of soil can provide energy to methylators (Gray and Hines, 2009), OM can bind with Hg(II) to form Hg-OM complexes to change the bioavailability of inorganic Hg (Yin et al., 2018), and a high OM content can maintain a low oxidation-reduction potential (Lambertsson and Nilsson, 2006). Jia et al. (2019) suggested that composition shifts of ammonia oxidizer communities could be well

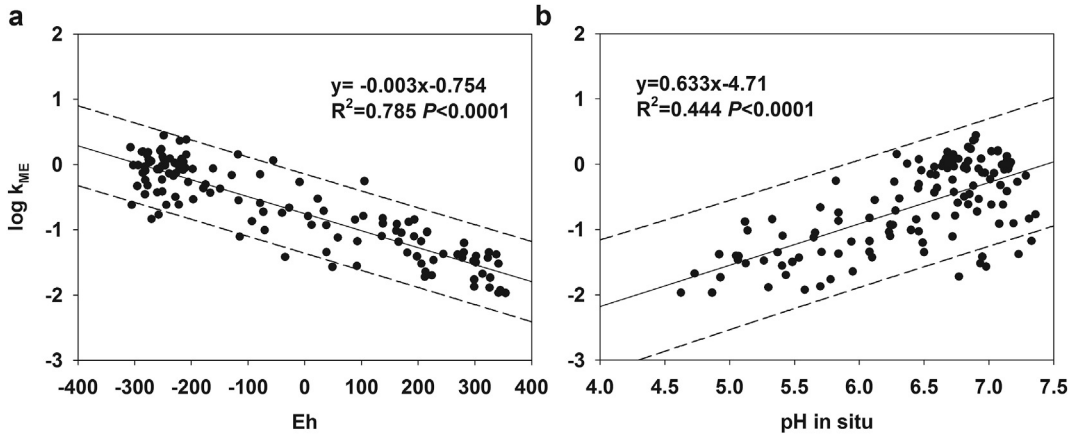


Fig. 3. The correlations between log k_{ME} and Eh (a), log k_{ME} and pH in situ (b). The solid lines indicate regression lines, while the dotted lines indicate 95% prediction intervals.

Table 1
Correlation matrix of b value and soil properties (n = 18).

	$b_{elovich}$	$b_{biconstant}$	pH	OM content	CEC	Selenium	Sulfate	Clay	Free FeOx	Free MnOx	Free AlOx	*hgcA* copies
$b_{elovich}$	1	0.481*	−0.204	0.479*	0.216	−0.089	−0.106	−0.079	0.175	0.281	0.321	−0.207
$b_{biconstant}$		1	−0.140	0.405	0.185	−0.182	−0.327	−0.027	0.213	0.398	0.362	−0.169

* correlation is significant at 0.05 level (2-tailed).

retained in both fresh and air-dried soils, despite the apparent decline in absolute abundance of ammonia oxidizers. The air-dried soil samples were used to measure *hgcA* gene copies (Fig. S1). No significant correlation was found between the *hgcA* gene copies in air-dried soils and the b values of the two kinetic equations.

3.2. Major controlling factors and predictive models of MeHg production in paddy soils

The optimum predictive models were obtained by minimizing the residual sum between predicted Hg methylation efficiencies and the measured values (Table 2). MeHg production might be well predicted by the in situ soil pH and Eh ($R^2 = 0.78$, Eq. (1) in Table 2).

Due to the fluctuations in soil pH and Eh after flooding and the inconvenience of measuring the in situ pH and Eh (Ding et al., 2019), the possibility of predicting MeHg production using soil properties, copies of the *hgcA* gene in air-dried soil samples and the flooding time was also investigated. The results clearly indicated that OM content was the most important soil property that controls the production of MeHg in paddy soils after flooding. The developed model also yielded good predictive performance ($R^2 = 0.679$, Eq. (2) in Table 2). The *hgcA* gene copies were used to develop the predictive model (Eq. (3) in Table 2), but the coefficient of determination ($R^2 = 0.642$) was not higher than that of the model using soil OM as the predictor. Therefore, the model using *hgcA* gene copies was not recommended. Introduction of the initial pH and sulfate content did not significantly improve the correlation performance (Eqs. 4 and 5 in Table 2). Other soil factors, such as CEC, clay, Se, free Fe, Mn, or Al oxide content, also failed to improve the prediction accuracy and therefore were excluded here. The accuracy of the predictive model was determined by plotting measured k_{ME} and the predicted values (Fig. 4). The majority of the predicted values were within the 95% prediction intervals. The root mean square error (RMSE) value was 2.38, and the regression line was close to the 1:1 line. Additionally, in Fig. 4, it was obvious that the coefficients of determination (R^2) were 0.783 and 0.679 for Eqs. (1) and (2), respectively, representing 78% and 68% of the variability of Hg methylation efficiency. Therefore, the developed predictive model was a reliable predictor of the k_{ME} of Hg methylation in paddy soils after flooding. The results suggested that the MeHg production of paddy soils after flooding can be predicted as a function of routinely measured soil properties and flooding time, a correlation that can be utilized to improve the understanding of the Hg methylation process and the management of Hg-contaminated paddy soils. The results in the present study provide insight into the extent and equilibrium time of MeHg production and the associated risk assessment of Hg-contaminated paddy soils. The result that the content of soil OM facilitated the production of MeHg could provide guidance for future remediation efforts, indicating that strengthening risk management and control is possible by prioritizing Hg-contaminated soil with high OM content. It was reported that the return of rice straw to the soil could enhance Hg

Table 2
The predictive models of Hg methylation process[a].

No.	Regression equation	R^2	*p*
1	$\log k_{ME} = -1.27 - 0.003Eh + 0.020pH_{insitu}$	0.783	<0.0001
2	$\log k_{ME} = (-2.08 + 0.259 \log OM) \times t^{0.203} + 0.908 \ln t$	0.679	<0.0001
3	$\log k_{ME} = (-2.40 + 0.225 \log hgcA) \times t^{0.201} + 0.900 \ln t$	0.642	<0.0001
4	$\log k_{ME} = (-1.98 - 0.025pH_{initial} + 0.288 \log OM) \times t^{0.204} + 0.913 \ln t$	0.683	<0.0001
5	$\log k_{ME} = (-2.24 - 0.085pH_{initial} + 0.338 \log OM + 0.345 \log SO_4^{2-}) \times t^{0.201} + 0.902 \ln t$	0.702	<0.0001

[a] $pH_{in\ situ}$ was the in situ soil pH, $pH_{initial}$, OM, SO_4^{2-} was the initial pH, content of organic matter and sulfate of air-dried soil samples, t was the flooding time (day).

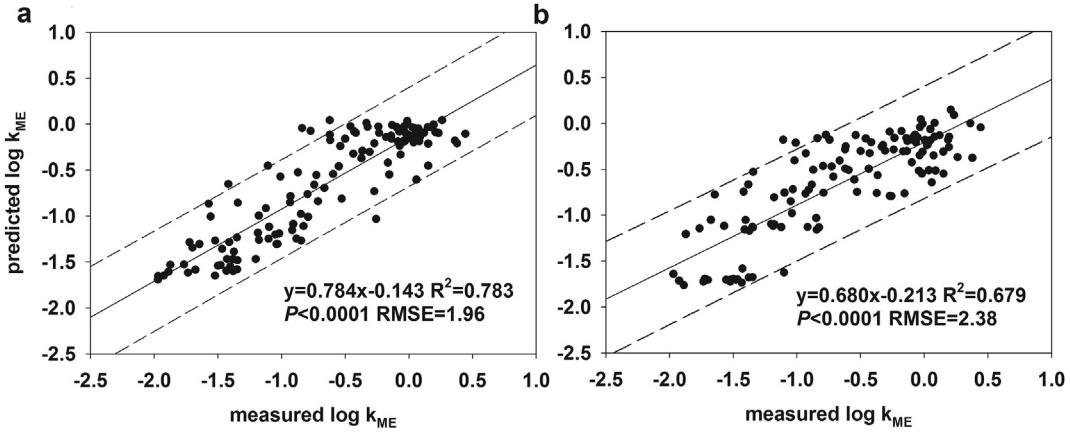


Fig. 4. Relationships between the measured log k_{ME} and the associated predicted values calculated from Eq. (1) (a) and Eq. (2) (b), respectively. The solid lines indicate regression lines, while the dotted lines indicate 95% prediction intervals.

methylation (Liu et al., 2016; Tang et al., 2018); however, the results of this study provided an underlying mechanism in which returning rice straw directly to the soil was not recommended in Hg-contaminated soils.

4. Conclusion

The methylation efficiency of exogenous inorganic Hg in a wide range of paddy soils increased with flooding time and reached a relatively steady state after flooding for 30 d. The Elovich equation could adequately describe the kinetic production of MeHg. MeHg production could be well predicted by the in situ soil pH and Eh of flooded soils, as well as the organic matter content of air-dried soil samples and flooding time. Future studies are needed to validate the developed predictive model under field conditions. Furthermore, soil microbe data and active Hg (Hg mass balance) analyses with flooding time, which were essential factors for demonstrating the methylation process and mechanism, should also be considered in future studies.

Conflict of Interest

The authors declared that they have no conflicts of interest to this work.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.envpol.2019.07.050.

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